

Science for Technology

Measuring Instruments (මිනුම් උපකරණ)

Short Notes 02

Tech Hub

01. Fundamental Physical Quantities (මූලික භෞතික රාශි)

Physical Quantity (රාශිය)	SI Unit & Symbol (ඒකකය)
Length (දිග)	m (Meter / මීටර)
Mass (ස්කන්ධය)	kg (Kilogram / කිලෝග්‍රෑම්)
Time (කාලය)	s (Second / තත්පර)
Electric Current (විද්‍යුත් ධාරාව)	A (Ampere / ඇම්පියර්)
Amount of Substance (පදාර්ථ ප්‍රමාණය)	mol (Mole / මවුල)
Thermodynamic Temp (උෂ්ණත්වය)	K (Kelvin / කෙල්වින්)
Luminous Intensity (දීප්ත තීව්‍රතාවය)	cd (Candela / කැන්ඩෙලා)

02. Measurement Precautions

- Select appropriate measuring instrument.
- Keep the measured surface clean.
- Do not measure moving or vibrating parts.
- Look directly perpendicular to avoid parallax error.
- Take measurements at standard room temperature.

03. Least Count (කුඩාම මිනුම)

The smallest value that can be measured accurately by an instrument.

- Meter Ruler: **0.1 cm (1 mm)**
- Vernier Calliper: **0.1 mm (0.01 cm)**
- Micrometer Screw Gauge: **0.01 mm**

04. Vernier Calliper Structure

[Diagram: Vernier Calliper - Main & Vernier Scale]

- **External Jaws:** Outer diameter/thickness
- **Internal Jaws:** Inner diameter of cavity
- **Depth Probe:** Depth of small holes

05. Vernier Calliper Errors

[Diagram: Positive & Negative Zero Error]

- **Positive Error:** Vernier 0 is right of Main 0 (Subtract error).
- **Negative Error:** Vernier 0 is left of Main 0 (Add error).

06. Micrometer Screw Gauge

[Diagram: Anvil, Spindle, Sleeve, Thimble, Ratchet]

- **Anvil & Spindle:** Holds measured object
- **Thimble/Ratchet:** Controls pressure/movement

07. Micrometer Least Count & Errors

Least Count = $\text{Pitch} / \text{Circular Divisions} = 0.5 \text{ mm} / 50 = 0.01 \text{ mm}$

- **No Zero Error:** Circular 0 aligns with sleeve line.
- **Positive Error:** Circular 0 below reference line.
- **Negative Error:** Circular 0 above reference line.

08. Triple Beam Balance (තෙදඬු තුලාව)

[Diagram: Pan, Beams, Riders, Zero Line]

- Least count: **0.1 g**
- Set riders to zero & balance pointer before measurement.

10. Error Calculation Formulas (දෝෂ ගණනය)

- Fractional Error (භාගික දෝෂය) = *Least Count / Measured Value*
 - Percentage Error (ප්‍රතිශත දෝෂය) = *(Least Count / Measured Value) × 100%*
- Note: An instrument is suitable if percentage error < 1%.*

09. Electronic Balance & Stopwatch

- **Electronic Balance:** Fast/accurate mass measurement; includes zero (tare) button. Affected by temperature & draft.
- **Stopwatch Least Count:** Analog = 0.1 s | Digital = 0.01 s